

12.9 Graphing Circles

Lesson Introduction

 Benchmark	MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.		 Learning Targets	- I can graph a circle and determine its key features when given an equation of a circle. - I can solve mathematical and real-world problems that are modeled with an equation of a circle.	
 Benchmark Coherence	Building On: N/A			Working Toward: N/A	
 Essential Questions	- What are the two key elements of a circle needed in order to graph the circle? - How can you graph a circle when the equation is written in general form?				
 Lesson Narrative	Throughout this lesson, students will be using key features of a circle to graph a circle on the coordinate plane. This lesson activates prior knowledge regarding domain and range, and requires students to write inequalities based on the equation of the circle. Students are given real-life scenarios and asked to answer questions based on the information given in the problems.				
 Component Summary	12.9.1 Warm-Up (~5 min.)	Notice and Wonder with U.S. Capitol Rotunda (SE p. 273)	12.9.3 Your Turn! (15 min.)	Independent practice graphing circles (SE p. 275)	
	12.9.2 Guided Practice (25 min.)	Graphing circles practice (SE p. 273)	12.9.4 Wrap-Up (~5 min.)	Error Analysis graphing circles (SE p. 277)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Domain - Range - Center - Radius		N/A		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may struggle with relating the key features of a circle to the context of the question. For example, students may struggle making connections in question 2 during 12.9.2 Guided Practice.	Students may have forgotten algebraic concepts such as how to find the domain, range, and writing inequalities. Consider small group instruction and differentiating routines as a primary driver of learning in this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 12.9.2 Guided Practice provides a structured and interactive opportunity for students to revise and refine both their ideas and their verbal and written output. The main idea of this routine is to have students think or write a response individually, and then use structured pairings for iterations of feedback and revisions, providing multiple opportunities to refine and clarify the response.	MA.K12.MTR.7.1: Real-World Contexts This lesson provides the opportunity for students to connect mathematical concepts to everyday experiences. Consider encouraging students to apply mathematics to real-world contexts, and support students as they validate conclusions by comparing them to the given situation.
 Differentiate Instruction	This lesson requires the student to be able to activate prior knowledge of domain, range, and writing inequalities. Consider reviewing these critical prerequisite concepts through both demonstration and models. For graphing and writing inequalities, there are two On-Ramp videos that can act as scaffolding to be used before or during the lesson. Even though this lesson focuses on the not-strict inequalities, having the students review the key differences of both will be helpful.	
Lesson Notes:		

12.9.1 Warm-Up: Notice and Wonder

Component Overview: Students will make observations and write down things they notice and wonder about the dome of the U.S. Capitol Building.



12.9.1 Warm-Up: Notice and Wonder

The dome of the U.S. Capitol Building is 288 feet (ft) in height and 96 ft in diameter. The dome covers the Rotunda, a room with a circular plan. The architect created an orchestration of architecture, sculpture, and painting in the U.S. Capitol Rotunda.



This component provides the opportunity for interdisciplinary studies. Consider challenging students by having them incorporate information about the U.S. Capital Building to lessons they have studied in their Civics or other Social Studies classes.

1. What do you notice?

Answers may vary.

I notice that the rotunda has concentric circles.

2. What do you wonder?

Answers may vary.

I wonder what is the diameter of the innermost circle.

12.9.2 Guided Practice: Graphing Circles

Component Overview: Students are working with the equation of a circle in general form, and are asked to identify the key features such as center, radius, domain, and range. This will lead students into making a connection between key features of a circle and the context of a real-world problem. Consider giving students space and opportunities for analyzing errors, engaging in math talks, and collaborating in order to orient students to the content and each other.



12.9.2 Guided Practice: Graphing Circles

1. The equation of circle W is shown.

$$x^2 + y^2 - 4x + 10y + 13 = 0$$

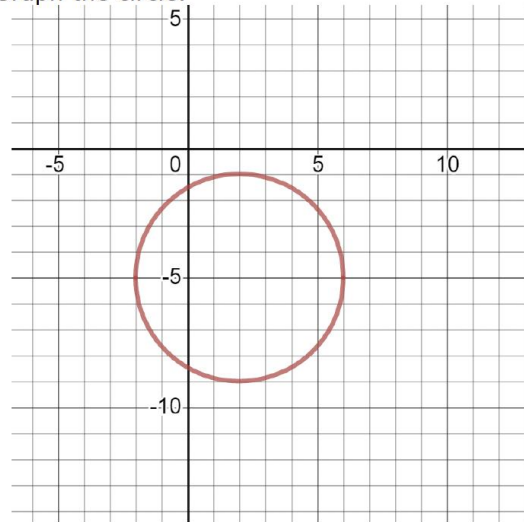
- a. Write the equation in center-radius form.

$$(x - 2)^2 + (y + 5)^2 = 16$$

- b. Complete the table.

Center	Radius
$(2, -5)$	4

- c. Graph the circle.



- d. Find the domain of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.

$$-2 \leq x \leq 6$$

- e. Find the range of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.

$$-9 \leq y \leq -1$$



Number Talk

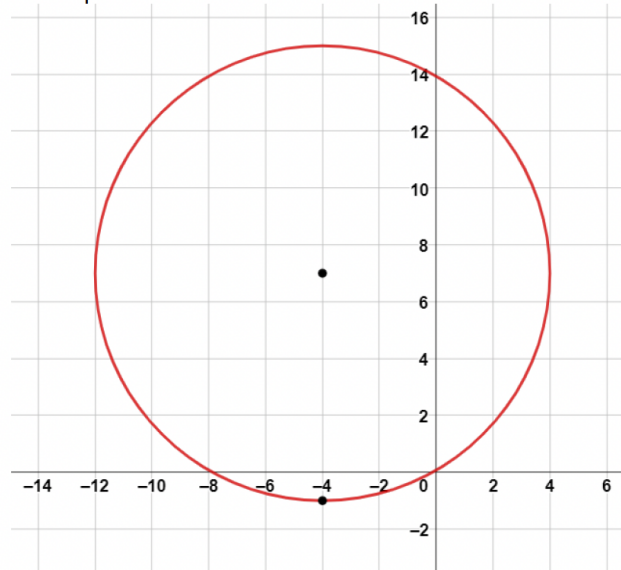
Display one question at a time. Give students think time before prompting a signal when they have an answer or strategy on solving each part. Students should find a partner based on shared signals and explain their findings.

Then, the teacher selects students to share different strategies for each part. ("Who thought about it in a different way?") Their explanations are recorded for all to see. Students might be pressed to provide more details about why they decided to approach a part of the question a certain way.

12.9.2 Guided Practice: Graphing Circles (continued)

2. Marcus wanted to order from a food delivery service and used the equation $(x + 4)^2 + (y - 7)^2 = 64$ of circle M to find food within his radius, in miles.

a. Graph circle M .



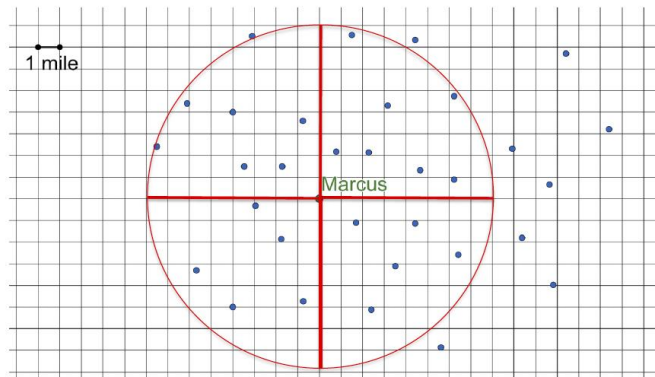
Monitor as students work through each question. Use inquiry to extend or scaffold student thinking when appropriate:

- What does the domain of a circle mean?
- What does the range of a circle mean?
- Why is the domain and range needed for Marcus' delivery?

- b. What does the center represent in the context?
The center represents the location of Marcus's house.
- c. What does the radius represent in the context?
The radius represents the distance in miles and area from which Marcus can order food.
- d. If the service charges \$3.25 for delivery for every four miles driven, what is the most that Marcus could be charged for the delivery?
\$6.50

12.9.2 Guided Practice: Graphing Circles (continued)

- e. Determine the domain of $(x + 4)^2 + (y - 7)^2 = 64$. Write the domain in set-builder notation.
 $\{x | -12 \leq x \leq 4\}$
- f. Determine the range of $(x + 4)^2 + (y - 7)^2 = 64$. Write the range in set-builder notation.
 $\{y | -1 \leq y \leq 15\}$
- g. The grid shows Marcus' location in green. The blue dots are the locations of restaurants that deliver using UberEats. Use the grid to demonstrate how Marcus could use the domain and range to determine which restaurants he can order from.



The domain and range will tell Marcus which restaurants he is able to place an order within his area.



Use higher-order thinking questions to informally measure concept mastery. These questions can be used as an alternative oral assessment for specific students. The following questions could be used:

- In your own words, explain the steps in converting an equation of a circle in center-radius form to general form. Create/produce an example to use in your explanation.*
- Explain how you could write the equation of a circle in center-radius and general forms if you were given only the endpoints of the diameter. Create/produce an example to use in your explanation.*
- Create/produce three truths and a lie about converting an equation of a circle from general form to center-radius form.*



What radius will Marcus need to be able to order from all the restaurants shown on the coordinate plane?

Why do you think food delivery services have a limit to where they deliver to?

12.9.3 Your Turn!

Component Overview: Students are given the opportunity to take what they learned in 12.9.2 Guided Practice and apply it independently. Probe students as they work, using assessing and critical thinking questions to check for understanding, to challenge students, and to help students who are struggling. Consider a peer review process for students to check each other's work as an added layer of support.



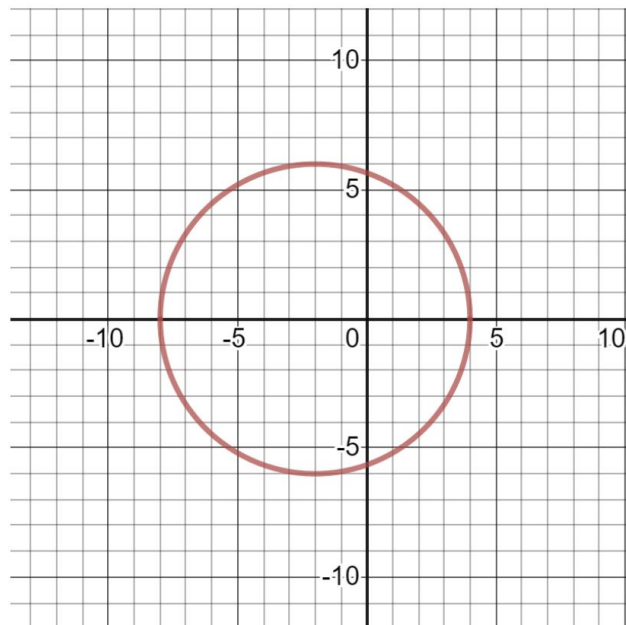
12.9.3 Your Turn!

1. The equation of circle R is $(x + 2)^2 + y^2 = 36$.

a. Complete the table.

Center	$(-2, 0)$
Radius	6
Domain	$-8 \leq x \leq 4$
Range	$-6 \leq y \leq 6$

b. Graph circle R .



Consider prompting questions for students who are unsure how to begin:

- How would you determine what information to use to start this question?
- Do you need to rewrite the equation before you can start? Why or why not?
- Do you need all the information provided? Why or why not?

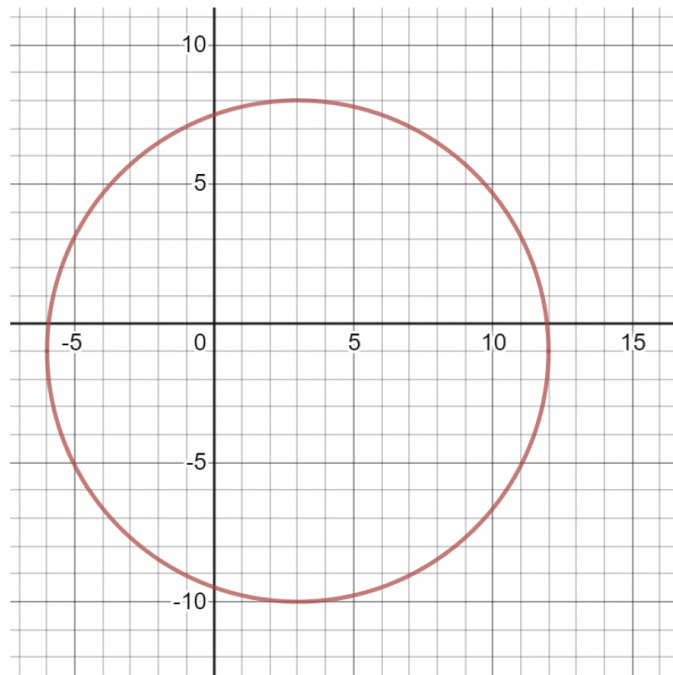
12.9.3 Your Turn! (continued)

2. Alyona used the equation $(x - 3)^2 + (y + 1) = 81$ to determine how far she could sit, in meters, from her Xbox and still use her wireless controller.

a. Complete the table.

Center	$(3, -1)$
Radius	9
Domain	$-6 \leq x \leq 12$
Range	$-10 \leq y \leq 8$

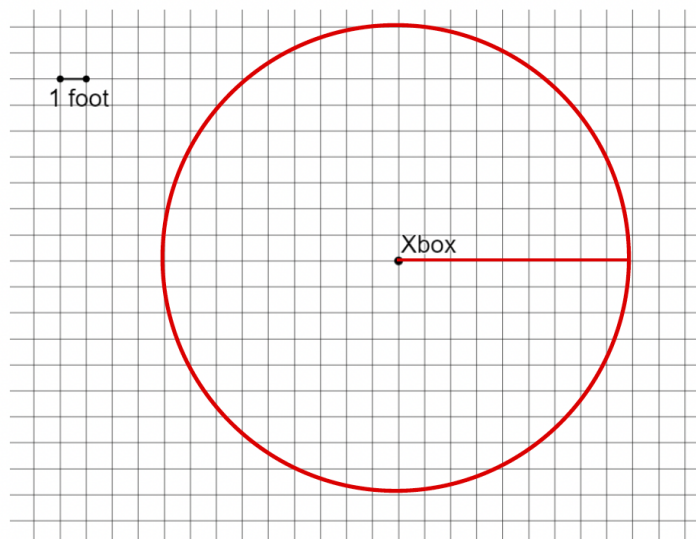
b. Graph $(x - 3)^2 + (y + 1) = 81$



For students who struggled during 12.9.2 Guided Practice, consider using this time to work with individual or small groups of students on interpreting an equation of a circle in real-world situations. Some students might not remember how to identify key features of a circle, and some might not be able to connect them to the real-world situation, so utilizing pairs or small groups will contribute toward a collaborative learning environment.

12.9.3 Your Turn! (continued)

- c. What does the center represent in the context?
The location of the Xbox
- d. What does the radius represent in the context?
How far Alyona can be from the Xbox and still use the wireless controller.
- e. The grid shows the location of Alyona's Xbox. Use the domain and range to show all of the locations at which Alyona can sit.



Consider challenging students who finish early to find the area Alyona's Xbox reaches.

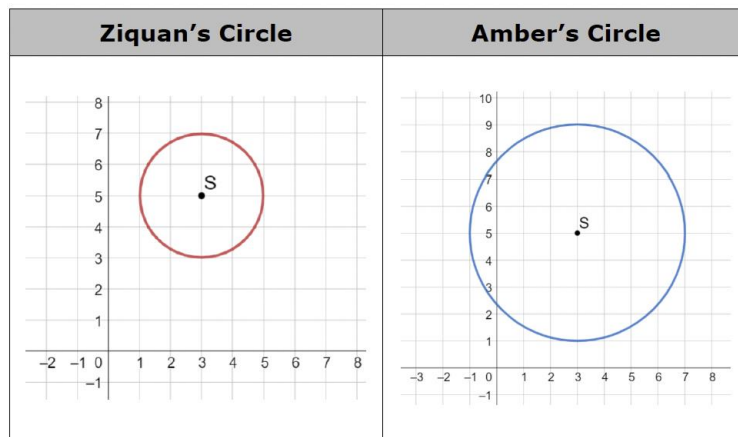
12.9.4 Wrap-Up: Error Analysis

Component Overview: 12.9.4 Wrap-Up assesses students' understanding of the key parts of a circle and how to graph a circle on a coordinate plane. Consider using all parts of this exercise as a formative assessment to gather data on students' progress from this lesson.



12.9.4 Wrap-Up: Error Analysis

1. Ziquan and Amber both graphed circle S , $(x - 3)^2 + (y - 5)^2 = 4$, but their circles looked different.



Visit the activities and Activity Builder on Desmos for a variety of engaging extension ideas for students to engage with following this lesson and unit.

- a. Who correctly graphed circle S ?

Ziquan

- b. Write a note to the person who graphed circle S incorrectly explaining their mistake.

Amber, the equation $(x - h)^2 + (y - k)^2 = r^2$ is the center-radius form of the equation of a circle. In the equation $(x - 3)^2 + (y - 5)^2 = 4$, $r^2 = 4$ so the radius is 2. You graphed a circle with a radius of 4.